

Introduction to University Mathematics

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Abstract

Notes for the University of Helsinki course MAT11001, taught in Finnish from 31 August to 17 October 2026.

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1 Course information

The official course information is available on the University of Helsinki course page.

2 Lecture notes

2.1 31.08.2026

The first lecture was mostly spent going over the course structure and other general admin topics. We did go over some preliminary definitions and examples, which will follow.

Definition 2.1. An integer $p > 1$ is *prime* if its only positive divisors are 1 and p .

Definition 2.2. The following basic sets of numbers will be used throughout:

$$\begin{aligned}\mathbb{N} &= \{0, 1, 2, \dots\} \\ \mathbb{Z} &= \{\dots, -2, -1, 0, 1, 2, \dots\} \\ \mathbb{Z}^+ &= \mathbb{N} \setminus \{0\} \\ \mathbb{Q} &= \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\} \\ \mathbb{R} &= \text{the set of real numbers} \\ \mathbb{C} &= \{a + bi : a, b \in \mathbb{R}\}, \quad i^2 = -1 \\ \mathbb{P} &= \{p \in \mathbb{Z} : p \text{ is prime}\}\end{aligned}$$

We use the following principle without proof.

Theorem 2.3 (Principle of mathematical induction). *Let $P(n)$ be a statement for each $n \in \mathbb{N}$. Suppose that*

1. $P(0)$ is true, and
2. for every $n \in \mathbb{N}$, $P(n)$ implies $P(n + 1)$.

Then $P(n)$ is true for every $n \in \mathbb{N}$.

Proof. Omitted. □

Example 2.4. We prove by induction that

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

for every integer $n \geq 1$.

For $n = 1$, the identity holds because

$$\sum_{k=1}^1 k = 1 = \frac{1(1+1)}{2}.$$

Now suppose that the identity holds for some integer $n \geq 1$. Then

$$\begin{aligned}\sum_{k=1}^{n+1} k &= \sum_{k=1}^n k + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{(n+1)(n+2)}{2}.\end{aligned}$$

Thus the identity also holds for $n + 1$, and hence for every integer $n \geq 1$. □

2.2 03.09.2026